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Next item

1. We want to use induction to prove the following formula: 1 / 1 point

$U_n = 1^3 + 2^3 + \ldots + n^3 = n^2(n+1)^2/4$

What is the basis step we need to verify?

- (x) $1^3 = 1^2(1+1)^2/4$
- ( ) $n^3 = n^2(n+1)^2/4$
- ( ) $1^3 + 2^3 = 2^2(2+1)^2/4$
- ( ) $U_n = 1^3 + 2^3 + \ldots n^3 = n^2(n+1)^2/4$

Nice work

Correct. In the basis step, we verify the formula for the initial value of $n = 1$ we have $U_1 = 1^3 = 1 = \frac{1^2(1+1)^2}{4} = \frac{4}{4} = 1.$

2. We want to use induction to prove the following formula: 1 / 1 point

$U_n = 2 + \ldots + n = (n(n+1) - 2)/2$

What is the inductive step we need to verify?

- ( ) $\exists$ n If $U_n = (n(n+1) - 2)/2$ then $U_{n+1} = (n+1(n+1+1) - 2)/2$
- (x) $\forall$ n If $U_n = (n(n+1) - 2)/2$ then $U_{n+1} = (n+1(n+1+1) - 2)/2$
- ( ) $\forall$ n If $U_n = (n(n+1) - 2)/2$ then $U_{n+2} = (n+2(n+2+1) - 2)/2$

Nice work

Correct. This is the correct inductive step. To prove the formula by induction, we need to show that if the formula holds for n, it also holds for $n + 1$.

3. We want to prove the formula $P(k) : 2^k < k!$ holds for all $k \geqslant 5$ 1 / 1 point

What is the basis step we need to verify?

- (x) $P(5)$
- ( ) $P(\infty)$
- ( ) $P(1)$
- ( ) $P(0)$

Nice work

Correct. This is the correct basis step. To use induction, we start with the smallest value $k = 5$.

4. We want to use induction to prove the following formula: 1 / 1 point

$\forall n \in N, P(n) : 21$ divides $4^{n+1} + 5^{2n-1}$

What is the inductive step we need to verify?

- ( ) $\forall n \in N$ if 21 divides $4^{n+1} + 5^{2n-1}$ is true, then 21 divides $4^{n+2} + 5^{2n-1}$ is true
- ( ) $\forall n \in N$ if $P(n+1)$ is true, then $P(1)$ is true
- (x) $\forall n \in N$, if 21 divides $4^{n+1} + 5^{2n-1}$ is true, then 21 divides $4^{n+2} + 5^{2n+1}$ is true
- ( ) $\exists n \in N$: if $P(n)$ is true, then $P(n+1)$ is true

Nice work

Correct. This is the correct inductive step. To prove the formula by induction, we need to show that if the formula holds for n, it also holds for $n + 1$.

5. After finishing the writing of an induction proof, we realise that the induction does not hold true for at least one integer. 1 / 1 point

What mistakes have we probably made while building the proof?

- (x) We proved the basis step without making sure that the inductive step is verified
- ( ) We have not made any mistake in the proof
- ( ) The basis step and the inductive step are both verified

Nice work

Correct. Both the basis step and the inductive step must be verified for the induction to hold.